

U.S. IT Hardware

The Quantum Leap: Winners and Losers in the Quantum future - Industry Expert Webinar Transcript and Takeaways



Mark C. Newman
+1 212 845 7822
mark.newman@bernsteinsg.com



April Li
+1 917 344 8339
april.li@bernsteinsg.com



Phoebe Sun
+1 917 344 8481
phoebe.sun@bernsteinsg.com

We hosted a webinar with a Quantum industry expert with 15 years experience in leading Quantum companies and large tech, as a follow-up to our recently published [Quantum Deep dive](#) note. This note is an edited transcript with key takeaways.

A key framework to understand the quantum landscape is the distinction between manufactured qubits and natural qubits. Manufactured qubits, such as superconducting, silicon spin, and topological, are engineered systems that benefit from fast speeds. However, they tend to suffer from higher noise and shorter coherence times. In contrast, natural qubits, including trapped ions and neutral atoms, are inherently identical with long coherence and strong noise resistance, while the main drawback being speed.

Currently, there are three leading modalities: superconducting (manufactured), trapped ions (natural) and neutral atoms (natural). Superconducting leads in scale, speed, and industrial maturity, supported by significant investment from large tech. Meanwhile, trapped ions and neutral atoms are advancing rapidly due to their superior qubit quality and natural ability to perform long-range interactions. Other modalities, such as spin qubits and photonic, are still earlier in development but show scaling potential, while topological represent a high-risk, high-reward approach that remains far from practical implementation.

The trade-off between speed and accuracy influences application fit. Problems that require repeated sampling to estimate real-valued outputs, such as chemistry simulations, materials science, favor fast systems like superconducting. Problems requiring a discrete correct answer, such as cryptography or combinatorial optimization, favor the higher fidelity of natural qubits. While IBM expects Quantum Advantage will be achieved this year, large-scale, fault-tolerant quantum computing is still many many years away. Progress depends on key milestones incl. development of logical qubits, logical gates, and scalable error-corrected systems.

Quantum computing is highly capital intensive. As a result, the dominant model will likely be cloud-based, rather than widespread ownership of machines. Early demand is driven by academia, government, and research fields with growing interest from industries like finance, logistics, and optimization.

Competitive dynamics favor large technology companies such as IBM and Google due to their access to capital, infrastructure, and talent. IBM, in particular, has a strong advantage in software and ecosystem like Qiskit, though its hardware leadership is increasingly challenged by higher-quality natural qubit systems. Startups are more likely to succeed through specialization, partnerships, or government-backed initiatives rather than competing head-on at full scale.

Overall, the industry is likely to evolve into a multi-modality ecosystem, where both manufactured and natural qubit coexist, each optimized for different applications, with hybrid quantum-classical computing emerging as the most practical path forward.

BERNSTEIN TICKER TABLE

Ticker	Rating	9 Jun 2026		Price Target	TTM Rel. Perf.	Adjusted EPS				Adjusted P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
AAPL (Apple)	O	USD	290.55	350.00	20.4%	USD	7.46	8.87	10.65	39.0	32.8	27.3
DELL (Dell)	O	USD	381.78	500.00	214.1%	USD	10.35	18.77	22.84	36.9	20.3	16.7
HPE (HPE)	M	USD	48.27	62.00	140.2%	USD	1.96	3.42	4.14	24.6	14.1	11.7
HPQ (HPQ)	M	USD	24.94	27.00	(25.4)%	USD	3.12	3.00	3.04	8.0	8.3	8.2
IBM (IBM)	M	USD	277.49	280.00	(22.5)%	USD	11.58	12.35	13.93	24.0	22.5	19.9
SNDK (SanDisk)	O	USD	1,646.54	1,700.00	3836.0%	USD	2.99	64.73	200.47	550.7	25.4	8.2
STX (Seagate)	O	USD	846.01	1,000.00	538.0%	USD	8.10	14.89	32.49	104.4	56.8	26.0
SMCI (SMCI)	M	USD	40.64	37.00	(28.3)%	USD	2.60	2.80	3.25	15.6	14.5	12.5
WDC (Western Digital)	O	USD	517.72	590.00	802.3%	USD	4.80	9.81	20.19	107.8	52.8	25.6
SPX			7,386.65									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

DELL estimate is Reported EPS; DELL, SMCI base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

DETAILS

We hosted Quantum industry expert with 15 years experience in leading Quantum companies and large tech, as a follow-up to our recently published [Quantum Deepdive](#) note. This note is an edited transcript with key takeaways.

Speaker Key:

Mark Newman = **MN**

Technology manager in the quantum computing industry = **Expert**

MN: Good morning, everyone. I'm Mark Newman, Bernstein's US IT Hardware Analyst. And today, we have the pleasure of hosting this webinar with a Technology Manager in the Quantum Computing Industry, who has 15 years experience in leading Quantum companies and large tech. At both companies, he worked on Quantum, so he has a wealth of knowledge and experience in quantum. And he is not biased because he is working for a non-profit currently. Expert, thank you very much for joining us. Anything else you'd like to add in terms of your background?

Expert: Yeah, absolutely. I've worked in the quantum computing industry and academia for the last 15 years or so. My career is sort of taking me up the whole stack that makes a viable quantum computer. I started off much more focused on hardware, even though I am a theoretical physicist by training, and then slowly over time moved to more system-level considerations, and then nowadays I work more on the software applications and algorithm side. My primary research expertise and focus for the last decade has been on characterization and benchmarking of quantum computers, so really understanding not only what they are capable of doing now, but what we can learn now to tell us about what they might be capable of in the near-term future and in the far-term future. So that's really been my focus.

MN: Great, thanks very much, Expert. And a reminder for everyone joining on the call, we do have a pigeonhole link where you can enter your own questions. I'm going to start off with a bunch of my own questions, and I'll refer to that later in the call to get some investor questions in as well. So let's get started. I'd like to start off talking about all the different, approaches, or what's referred to as the modalities for quantum. So, you have a lot of experience here, in quantum. You're neutral, as you said, working at a non-profit currently. If you had to rank the six modalities, can we talk about each of the six modalities and talk about them in terms of which ones you think have the best chance of success going forwards? I'll just list them all — superconducting, trapped ion, neutral atom, silicon spin, photonics, and topological.

Expert: Yeah, absolutely. So, I think, before I go into that, I want to put in a few caveat statements. Quantum computing as an industry is not necessarily following the path that classical computing and traditional computing has, in the sense that when you look at the early, early, early days of classical computing, when we had vacuum tubes and these sort of massive-scale systems, the industry was really revolutionized by the transistor. And it sort of unified around this single architecture that was going to be the one moving forward. That could happen in quantum computing, absolutely, and I think some of the modalities you mentioned there, even some that aren't on that list, that are even less solidified now, could have that kind of impact. But at the moment, we don't anticipate that necessarily in the next 10 or 15 years. It's one of these situations where an absolute revolution in science and in physics could cause this amalgamation or homogenization of the architecture. But right now, in my view, there's no clear winner in terms of the long-term bet. I think each architecture has their advantages and disadvantages, and it's a hard engineering and a hard science problem to overcome the roadblocks that they each have to getting to a large-scale quantum computer. There certainly are ones that are leading right now and ahead of the game. But these sort of revolutions do even happen on a small scale. Like, I'm gonna place neutral atoms somewhere, and if we had this call two years ago, they would have been in a completely different location on the scale. So, these kind of things can happen, just keep that in mind, that information needs to be refreshed periodically.

Expert: I like to break things into two buckets, into manufactured qubits and natural qubits. So manufactured qubits are superconducting qubits, the spin qubits, and the topological. The natural qubits are the trapped ions and neutral atoms. And then, kind of in the middle is photonic. Photonic qubits are "natural," because they're light. But they require manufactured waveguides, manufactured substrates to live in. Now, this is actually also true of trapped ions, trapped neutral atoms, they have manufactured traps that they live in. So, the lines are all a little bit blurry here, right? But the reason this distinction is important is because it really impacts the quality of the individual qubits themselves.

With the natural ones, where you would see the trapped ions, the neutral atoms, and the photonic qubits, these qubits themselves have really good properties. One, they're consistent, every single one of them is the same in the world. And that makes a big difference about designing your control architecture, so the higher level stuff in the stack that controls the quantum is much more reliable, right? If you put 50 atoms in a trap, you know that all 50 are more or less the same to levels of accuracy that you don't really care about. Because they're natural the parts of their system that we use to store information also have

very natural properties. They're naturally long-lived, so they have good coherence times, which means that we can store the information for a long time. They're naturally robust to error, because again, these are systems that exist in the isolated world, independent of us creating them. So they have a lot of nice properties like that. The flip side is that all those properties that make them nice make them very hard to control. The control is very slow, so you think about the operation rate of these natural qubits, the ions and the atoms, it's at the... you can repeat a computation at most at a 1Hz repetition rate, so once per second. Some of this is engineering challenges, and these groups are working very hard to overcome them, right? But a lot of it is just the physical, natural operational speed of these devices is much slower. I'll touch on that in a bit.

Expert: By contrast, the manufactured qubits, they have much faster operation rates, because we are making that to be the case, right? So they operate at kilohertz, or even megahertz repetition rates, where we can repeat computations at a much faster timescale. The natural clock speed of the operations are also much faster, because again, we've designed the system to be the way we want. Where they fail, so where the superconductors, the spin qubits, and the Majoranas suffer, is that they, don't have as nice natural properties, with the exception of the Majoranas, which I'll get to in a second. But the superconducting and spin qubits are naturally noisier, they're not as consistent. When you make two superconducting qubits, for instance, even if you try to make them the same, you're not going to. There's a roughly 10% spread in fabrication, typically. That means your control systems have to be a lot more robust and work in a much larger bandwidth because they have to adapt naturally to the spread and the qubit properties. They don't have as good coherence times either. Because they're these macroscopic manufactured systems, they're much more susceptible to noise in the environment, and they have to be much more carefully isolated. And all of this means that typically the manufactured qubits are going to have worse properties. They're going to have worse error rates, as we call them, per operation. They're going to have worse coherence times, and the benefit for them is really the speed versus quality of the natural qubits.

Now, if we go on to dive into the details, each one in particular, the way I view it right now is sort of a three-way tie at the top, and it's hard to really say which one is leading, because it really depends on what you're looking for. And that three-way tie is between superconducting qubits, trapped ions, and neutral atoms. Two years ago, I would have put neutral atoms probably at the bottom of the whole list, right? In the last 3 years, there's been really revolutionary work out of the Harvard and MIT collaboration that has a spin-out company, QuEra, and they've really pushed the field. But it's important to caveat that they're so far the only ones who can get this good performance out of neutral atoms, whereas we see multiple trapped ion groups and multiple superconducting groups in industry and academia demonstrating good performance overall.

The superconductors win on speed, in the sense that if you want to do many, many repetitions of a circuit of a computation, which is something you naturally need to do for quantum computing - you can do that orders of magnitude, factors of a thousand, faster than anywhere else on them. They also have one of the largest scale computers, hundreds of qubits, and their quality has improved drastically in the last 5 years. So the quality of the gates is good, the reproducibility and robustness of the gates is very good. They're certainly, very ahead right now overall. So they're just maybe the 1A, right? The problem with them is the long-term vision is a little bit more murky. Because they're manufactured planar devices that exist on a flat plane, it's very hard to engineer interactions that aren't nearest neighbor, aren't between two qubits that are next to each other. And what we believe is that these longer-range interactions are necessary to do better error correction codes that will lower all the error rates to be where we can reach this fault-tolerant logical quantum computer that is going to be able to solve the applications we really care about. So there's tremendous savings in the number of physical qubits you need to make one logical qubit if you can enable these long-range interactions. That naturally just doesn't happen in superconducting qubits. Groups like the big industrial people like IBM and Google are working on finding ways to be able to do that. And I actually just saw this startup in Europe, IQM, has just released today their roadmap for that, for how they will enable these long-range interactions. So that's where I think things get a little murky. It's a lot of, to be frank, world-first demonstrations anywhere, even outside of academia, that has to be done for these groups to achieve that.

But, preliminary results that we have seen publicly look good, so they are on their pathway, but it is a big engineering and technical challenge. The ions and atoms naturally can do this, either by moving the ions around, or moving the atoms around as well, which are slightly different technologies to do that. So they can enable these long-range interactions, these multi-qubit connections, very naturally. And we'll see, probably in the next couple years, very quickly demonstrations from these groups, of those logical or early fault-tolerant demonstrations. But it's going to be hard for them to overcome this, the slowness of their sampling rates. So for applications like chemistry, material science that require estimation of a real number, where you need, you know, to some precision that's given by how many times you can repeat the computation. They're going to struggle, because the computations are going to take a thousand times longer. It's really a trade-off here, but there are applications that don't need that, like factoring, that just need the answer once, right? As long as you get it. So, you know, it really is application-dependent which one is going to be a long-term winner.

Expert: Spin qubits, photonics are at a much earlier stage, even though they've been explored for just as long, and photonic quantum computing was one of the first proposals. Photonic quantum computing has always, struggled with being able to interact photons. They naturally don't interact, so doing gates between the qubits is very hard, and that's still a problem. You have some companies that are exploring it, they have proposals, but we don't really see a lot of actual demonstrations coming out of those, so it's hard to judge where they're really at. Similarly, spin qubits are notoriously noisy and dirty and just difficult to manage. Once you can solve the problem of how do you make a high-quality spin qubit, how you can make interaction between two of them, it's very easy to scale because it's naturally semiconductor tech. We haven't gotten to that stage yet where we can actually leverage the scaling advantages from semiconductor tech. So I've put those at a joint, sort of, 4-5 area of that list. A lot of groups are looking at these now, especially in industry. You have Photonic Inc in Canada, you have Quandela in France, looking at this hybrid mode of both spin and photonics, using spins as the logical memory but doing all interactions via flying photonic qubits. No one has actually demonstrated that yet, so again, this is one of these long-term visions, but it's a way to kind of overcome the worst parts of both architectures and utilize the best parts of the two: use the naturally high coherence time of a spin, but its inability to interact with anything, then use the photons as information carriers to do that long-range interaction between spins. It's a long-term bet, we don't know if it'll pay off, but it's what some of the people are pivoting to.

Expert: And then the last, to be honest in my view right now, is the Majoranas. Not the last because of its promise. Topological qubits like Majoranas at Microsoft are doing work, I mean, they solve a lot of problems naturally, because they're naturally protected from noise and from error. But, they're very, very far away from being a real demonstration. No one's been able to demonstrate a single qubit, despite what you might see in popular press when you boil down to the technical demonstrations. We're at the stage of demonstrating the possibility of a classical logical information carrier, so a bit existing in the system. Topological physics is, of course, very hard to demonstrate. It requires these heterogeneous structures of superconductors and semiconductors that are notoriously difficult to manufacture, so there's a lot of outstanding problems. So I think they're one of those ones that are poised for revolution. If they can demonstrate this and it works, they will very quickly catch up with the rest of the field. But it is a big "if?", and it is, it would be a Nobel Prize-level kind of demonstration.

MN: Okay, great. So, in conclusion, top three being superconducting, trapped ion, and neutral atom, those are the top three. I wanted to just circle back on something you said that was interesting. You were saying that certain types of applications may be better served by superconducting because of speed, versus some which may be better served by something like trapped ion or neutral atom. Could you just double-click on that in terms of which type of application will be better for each type?

Expert: Absolutely, so I'll give a very technical description, but try to make it understandable, and then...

MN: Not too technical.

Expert: Not too technical. Yeah, so basically, well, I think it's interesting because it kind of ties into the way people think about modern AI and generative learning. If the application you need wants a real number, so like an energy from a chemical molecule, or a reaction rate from a surface material. Those are going to require a lot of shots because you're going to need to estimate that real number to an accuracy that is given by 1 over the number of repetitions, the number of shots. So that's where you need something that's fast. Those kinds of applications need many, many samples, and you can often get away with not being so accurate because there's techniques you can do to try to minimize the impact of your errors, the errors in your computation. You're competing against classical algorithms that are by their nature, also approximate. It's okay to not always be correct if you can get a lot of data.

Expert: On the flip side, applications that don't want real numbers but a distribution of outcomes, so that's things like factoring, where you're just looking for the factors, right? You're gonna get some distribution of numbers that could be the factor, or pairs of numbers, and you want the one that is the pair. And it's gonna have the highest probability, hopefully, that you get that right number. Or things like optimization problems, where you're looking for the optimal path for a salesman to take, even though that's too hard a problem, but things like that, where there's a discrete answer that comes out at the end that has a binary representation. Those are where you want high accuracy because you want the probability of getting the right answer to be really high. But you don't actually need to take a lot of shots, because once you get the right answer, you're done.

MN: What's an example of what you said for factoring? What's a real-life example there?

Expert: Factoring is useful for encryption breaking, so if you want to break RSA or elliptic curve cryptography, Shor's algorithm, which is the one of the first quantum algorithms developed, will break that by factoring the co-primes, products of two prime numbers, that go into the public key. For RSA, for instance, you want to make sure the algorithm, even the noisy version, has a high probability of giving you that right answer immediately. That's where you want the maximum quality you can get, because once you've run it once, gotten the answer, run it twice, however small number of times it takes, once you get the answer, you're

done. You don't ever need to run it again. So that's where the quality over speed is important. So cracking codes, these types of things, trapped ion and neutral atom are better. But for chemistry, materials science, you need more shots and more speed, so superconducting is potentially better. It's not a hard and fast, right? There are ways to use the sampling version for material science, for discovery science, for chemistry, and there's machine learning applications, for instance, that don't need real numbers but quality samples? It's not hard and fast, but that's sort of, in my view, like a broad picture categorization.

MN: And what about biology? Like trying to understand how certain drugs interact with proteins? Would that be more like chemistry?

Expert: It is, in one avenue. There's also ways to think of it like genetic learning, genetic algorithms where you can think of it more as samples. You want high-quality samples, and then you can... and this is where I will lead into some combination of classical and quantum, where with those high-quality samples, you can feed those into classical algorithms that do the bulk of the optimization, and what the quantum is doing is preparing these high-quality initial sample states that they use. Obviously, people are trying to get both... all the applications to fit both cases, right? So that's where the field is working hard to find cross-cutting ways to solve problems.

MN: So, potentially, if there's a breakthrough in one or another of these modalities, then one could be winner-takes-all, but at the moment it seems more likely that you're gonna get multiple modalities win out for different applications.

Expert: Absolutely, yeah, I think so.

MN: Okay, and which metric should we actually be looking at when we're comparing progress between companies here? Because there's a lot of numbers that are thrown out by some of these competitors, IBM, etc. So, is it physical qubit count? Is it two-qubit gate fidelity? Is it coherence time? You know, what are the ones that we should be looking at?

Expert: Yeah, so a physical qubit count isn't in and of itself really useful. I would say this is an evolving problem right now. Like, this is what my research is on, right? People are working on ways to better quantify things. A concept like what people would call an active volume, so it's how many qubits can be actively used at once in a computation, is more useful. For a manufactured qubit, it's very easy to build a thousand qubits, but you're not going to be able to control them all, or use them all. For instance, for IBM devices, you want to see that you can actually create a large quantum state across all the qubits, so you want to be able to run an algorithm that has all of them involved. I think physical qubit count is useful, obviously, because you need a big enough system to do operations, to do something useful or interesting, but it has to be actively available or useful. When it comes to gate fidelities, the field has kind of moved away from the single qubit gate fidelity, et cetera, being that useful a metric. What we really look at now is the two-qubit gate fidelity, not in isolation, but in the context of a larger operation. So, running a simultaneous operation on 15 qubits, on 20 qubits, on 100 qubits. And the reason for that is all of the architectures suffer from what people call crosstalk, which is when a gate over here unintentionally impacts a gate over there, unintentionally changing the way it performs. But in reality, no algorithm happens gate by gate by gate by gate by gate. You're gonna use as much of the real estate as you can simultaneously to minimize computation time, right? So that's where you want to look at metrics. I mean, I developed these ones at IBM, Layer Fidelity and others, but they're applied broadly, looking at simultaneous performance that plays into some of the other architectures as well.

Expert: The superconducting qubits can do a lot of gates at once, not all as high quality, a lot more impact of this crosstalk. Neutral atoms, trapped ions, can't do as many at the same time. What they do is they move ions around, so they have to do subsets of these operations, but they're all much higher quality. So these kind of metrics of, for example, how if you did all 10 gates at once what the performance would be; how they're moving the ions around, how they're moving the atoms around; how they're handling those transformations, scheduling the different operations sequentially. All those things impact performance of algorithms. These sort of system-scale metrics that can be boiled down to a single number are useful because they give you an overall idea of quality. But as a characterization person, I have to say, the unfortunate reality is all of these metrics are designed to be average, right? It's how good on average would you do? So your actual application will differ. And that's the hardest problem right now in characterization, it's figuring out a better way to connect to specific application performance.

MN: Great, thanks, and which modality would you say, is the most underappreciated right now? Which has most momentum, potentially one that the market is not taking enough notice of?

Expert: That's a good question. I would say the market's taken very well to most of them. I would say there's some more niche approaches to superconducting qubits that do what are called bosonic encodings, so rather than encode the information in a superconducting qubit, they encode it in a microwave resonator, so in light but at the microwave frequency, and they use the qubits for control. And you have companies like Alice & Bob in France, Nord Quantique in Canada, Quantum Circuits, who just

got bought out by D-Wave in the US, they were doing this. And that has a lot of promise because it gives you high coherence time, and it gives you naturally error-robust properties. It's a bit hard to control. I think maybe those are still a little bit under-appreciated for their potential in the near-ish term. They may not be the longest long-term bet, but on the middle ground scale, they might be able to demonstrate some useful devices quicker, even if then they run into scaling problems of their own.

MN: Okay, thanks very much. I'd like to move on to the next topic, which is around the timeline to quantum supremacy. Can you walk through a base case for timeline in terms of when we get to this kind of desirable quantum advantage on really meaningful, commercially relevant problems?

Expert: Yeah, I think there's two or maybe three gates. There's quantum advantage on anything that robustly stands the test of time over more than two months. There's quantum advantage on something commercially viable, and I would say that might be a business or research application. Large-scale quantum advantage is the third stage, where we're solving very hard problems routinely. Depending on who you ask, you're gonna get different answers. For instance, IBM claims that they'll get that first kind of quantum advantage, maybe even the second one, this year. On the flip side, the NIST estimates it's coming in or around 2035 in their recommendation for adopting post-quantum cryptography, cryptography that's resilient to quantum attacks. Essentially they're thinking that the possibility exists by 2035 that you could have a cryptographically relevant quantum computer that can run Shor's algorithm. That's a very, very large scale and complicated machine. I think all of this should be taken with a grain of salt. My view, we're probably looking at a 15-year time frame before we're at the scale where we're doing Shor's algorithm kind of problems. Maybe 20 years. Even here, I feel like I'm being a little bit optimistic. And there's really that question of what happens in the in-between. I personally find timelining very hard right now because we're at the cusp of a transition from purely physical devices, where we're operating just on the bare qubits, to these early fault-tolerant logical devices, where we're starting to encode qubits together to do redundant encoding to get higher quality logical qubits. And I'm worried what we don't know right now is really the performance of those initial devices, because right now, first groups are starting to demonstrate it in the industry. If that works well, I think we can see that I would be comfortable saying is limited quantum advantage in some specific cases in the next 5 years, probably in research science but not so much in commercially applicable things like financial market prediction or fraud detection. One thing I do want to point out, though, is we always think about advantage as better, as faster, as cheaper, as less energy. But advantage can also mean different things, and what we're really starting to see that now is that quantum computers, even noisy ones, can unlock different ways of preprocessing information, which we can then hand off to a classical computer to do the actual processing. I feel like in the next 5 years, you're going to see that kind of advantage where the quantum computer is one part of an overall HPC system, even though it's not doing the heavy lifting. It's just doing one small part that gives a difference in the outcome. It might not be cheaper, it might not be faster, but it could be different in a way that is useful to the application. And I think in some of the things that do with outlier detection, like fraud detection, that could be useful because they potentially change the way we encode data, learn data, and separate classes of data. I think we're almost there, to be honest, but none of this touches on just the accessibility of all of this, and whether or not we're going to be able to use it realistically in a deployed environment. The existing cloud infrastructures are slow, so there are lots of other implementation questions that come into meeting real value. But in terms of the pen and paper, I can say that this is doing something different and slightly better, or meets my requirements better. I would say in the next 5 years, we should hopefully see that.

MN: And any more specifics in terms of concrete signposts that we can look for between now and this kind of more meaningful quantum supremacy?

Expert: I think a lot of those signposts are going to be very "technical" in nature because they're going to be about demonstration of error correction. I don't really take application demonstrations as necessarily the best signposts, because it's very hard to do cross-platform comparison, or cross-application comparison in quantum computing, because the primitives that underlie algorithms are very different. But if you're talking more broadly you've got to look at where the hardware is going. When companies announce, "Okay, we have a logical qubit, and it has this fidelity," that's the first signpost. Great. They're halfway there. When they say, "Okay, we have a logical gate," now that's a big step. "Now we can do two encoded, protected operations between two different logical qubits." That's the next big step. And then you're looking for, "Okay, now we have tiled a system, we have 10 logical qubits, and we can do gates between all of them." So those technology signposts are pointing along that long-term pathway. Because the tricky part with these long-term algorithms, like Shor's algorithm, like some of the ones that do very high-performance chemistry and material science, is that there are no in-between. It's either all or nothing. What you're looking for is the building blocks of finally being able to do that algorithm coming into play. And to be honest, you're going to see a lot of roadblock signposts that have nothing to do with quantum. When IBM or Google or whoever, or someone super connected makes the announcement of CryoCMOS or SFQ control natively in their hardware, that's a huge signpost, because now they've removed the many hundreds of millions of dollars of classical control electronics that sit outside their dilution refrigerator. They've miniaturized it and put it in the fridge, and that unlocks huge scaling potential and performance improvements. When the trapped ions people, for instance, say they've optimized ion routing, which they're doing right now, that's another huge sign

post, because they're saying we can now move these ions around in a way that minimizes the transfer time and can enable faster logical operations. Or when they say something like, "We've improved the reloading time of a neutral atom processor by an order of magnitude." That's an order of magnitude faster shots. So, these are the kind of things you're looking for that honestly have nothing to do with sometimes even the qubits, but it's all those supporting technologies, it's the logic, it's the encoding. And I admit, it's hard because these are technical things, and people are moving away from reporting these things, but they're still publicly made available. You just have to kind of dig maybe to a second layer to find this information coming out.

MN: Yeah, that's really helpful. And I wanted to talk a bit more on the demand side. Who are the early buyers here, potentially markets that may be interested to pay for this, end markets, but also in terms of research departments, universities, governments, etc? Could you talk about that, in terms of who matters, and how that may develop over the next 5–10 years?

Expert: Absolutely. want to open up by saying, I was at a talk recently from Xanadu, who's a Canadian quantum computing company, they build photonic computers. They made a statement, where they were saying that, "What's it going to take to build a fault-tolerant computer? How much is it gonna cost? It's going to cost a billion dollars." Just to manufacture it. So, you can imagine how much they're going to sell it for, right? And similarly, I recently did some calculations. What would it take to operate a superconducting quantum computer to do one instance of cryptography code breaking? So break one key. It's gonna cost \$100,000 in electricity just to run it once. The cost of these large-scale applications is tremendous. So that's why, when you think about who's gonna buy the computers, it's very few people. It's going to be this model of cloud access. I think that's what you're getting for, but I wanted to make that statement first, that the computers themselves are very expensive. It's really an HPC comparison we're making here. So who's going to be adopting the cloud services? We've seen a lot of adoption in what I call discovery science. So that's academic groups in universities, in research industry, who are working in chemistry, material science, biology. There's a growing interest in high-energy physics, and even questions of cosmology. So those are a lot of the early adopters. But at the same time, we're seeing increased interest from financial institutions, people who want to do market predictions, people who want to do fraud detection, like I mentioned, people who want to do analysis of financial data. And we're seeing growing interest in, logistics and operations people, so routing for vehicles, optimizing manufacturing schedules. The core, sort of, element of all of this is it's hard math problems to solve. The original market was going was always going to be for cryptography, for breaking RSA, for breaking elliptic curve cryptography. Because of the computers those require, those are those long-term looking things. I think we're hoping that in the near term, we're gonna find more applications in logistics, in operations research, but none of these, unfortunately, and that includes some of the chemistry and biology ones, come with guarantees of advantage. So it becomes this heuristic problem of comparing against existing classical methods.

MN: Sound good. I'm gonna switch to a slightly different topic in terms of... looking at, some of the players, and particularly big tech versus startups or smaller pure plays. Do you think this ends up looking like a capital intensity game that only the big companies, like IBM, can fund, the path to fault tolerance? Or do you think some of these more focused pure plays can win on a technology lead?

Expert: I would say, honestly, my opinion is that it's only big tech that can get to fault tolerance. And I want to caveat that by saying that that doesn't mean that there isn't value in what the the startups are doing, even necessarily in the near term. I think, even more than capital, it's the ability to leverage that capital efficiently. You look at a company like IBM, for instance. I use them as an example because I used to work for them. The existing semiconductor manufacture chain, the existing technology experience there, allows them to move much faster and leverage the money that they do have much quicker to make advancements. Whereas you take a startup and, like, let's take Rigetti, for example, in the US. They had to build their own fabrication foundry. They spent tens of millions of dollars of doing it. Was that the best route forward? I mean, maybe, we don't know. Could they have leveraged fabrication partners? Yes, but there wasn't really anyone existing. So, they had a lot of capital, and they did it. But it set them back for sure. There's also an issue of hiring in the community where it's hard for emerging players to hire good talent. Because the long-term bet is so far off, you can't entice people with the normal sort of startup incentives, right. Of, "Here, I'll give you X percent of the company, and you'll be a millionaire one day if this works out." What works out might be 20 years from now, so it's a harder incentive for people, versus you go to Google and they pay you 3X. So, that's definitely an issue that a lot of the startups are having, and that also ties into them not being able to leverage the capital appropriately, because they don't have the people with the experience, with the know-how to scale out a big quantum computer. You know, the average is tremendous. Google has a thousand people working on quantum. IBM has the same number of people. So it's going to be hard for a company, a small startup with 30 people, to be able to get to that scale quicker and quickly enough, and Smartly thought out enough to really be able to build something as large as a fault-tolerant quantum computer.

MN: But in that case, what do you think for some of these smaller build plays, what do you think is the end game? Do you think, acquisition targets for the bigger players? Or do you think that there's also this concept of building a foundry for... I think, I mean, Infineon, makes, trapped ion chips for IonQ, I believe, for example. I believe the US government has been trying to fund

some kind of foundry. Do you think that could help the smaller players win if there's a meaningful foundry that can facilitate the manufacturing side?

Expert: I think absolutely. I think if there was a meaningful foundry for enough of the architectures, and you could go to more of this fabless model, it would enable a lot more, quicker R&D, because the smaller companies could focus on control electronics, could focus on architectural questions, system design, they wouldn't have to worry about building qubits as much. Or, I think acquisition targets is definitely one avenue that some of the startups are looking at. You know, something that's coming up more and more in the context of people outside the U.S. is sovereign capture. A smaller country saying, "This is our quantum computing company, we are supporting them," right? You see that with Indian players, you see that with France as well with Alice and Bob, and others where they want a domestic, sovereign quantum computer. So those governments are investing a lot of money in that, which also gives them a bit more of a runtime to slowly build things up. I mean, Xanadu has done a great job of this, starting from small to several hundred people now. Their technology development aside, they've at least slowly built themselves into this robust model over the last 10 years. And that's because of a lot of government support. I think when we've seen this try to happen purely through investment in venture capital, it's been a bit trickier. These issues with having to fire half their staff, and similarly with D-Wave. You really need an investor who's willing to take that long-term bid so you can overcome those issues of capital utilization that will happen if you try to scale too quickly.

MN: And can we talk a bit about the business model and distribution channels? So, you know, there's one approach, which is these cloud providers, so AWS Braket, Azure Quantum, IBM Cloud where they're essentially renting to quantum computing, versus, in some cases, companies are selling quantum computers to customers. What do you think happens longer term, and do you think that cloud distribution channel matters, and does that let incumbents commoditize some of the hardware startups, regardless of who has the best qubits?

Expert: I think absolutely long-term, for large-scale fault-tolerant computers, cloud distribution is the only way that makes sense. Because these are data center scale and HPC-scale objects, I think what we will see, especially in the near term, is an emergence of deployment model where people are selling quantum accelerators for on-premises devices. So right now, your IBMs, your IQMs, your Rigettis of the world are selling Quantum computers that are meant to be general purpose. For on-premises deployment, they're small. There's definitely a value there in workforce development, understanding the hardware, etc. But in practice, they're no more useful than accessing the cloud servers for application research and application development. But what I think we'll see now as we enter this early fault-tolerant era is custom-made accelerators that can be put on premises that are much cheaper, but are application-specific. A lot of this comes requires us identifying what the applications are and building the right architecture to solve them, but I think that will be an emerging market that some of the smaller players who can't scale up to the full fault-tolerant device, the long-term vision can capture. But I think the long-term vision, things like AWS Braket in my view, the longer-term vision for these general-purpose access models, where you want to do a Shor's algorithm, or an arbitrary computation, is you're going to have a quantum computer that exists in a data center, and that's going to be it. You're not going to be building these in every office. So I think you're right that that will enable the capture of a lot of the existing markets. We see this already with AWS. They have no quantum computers of their own available, but they host everyone else's.

MN: Right.

Expert: And they actually have spent a lot of time working on good software interface to enable use. Because this is a lot of what I do now, I will say that getting the most out of a cloud computer is a very hard problem because we haven't had 30 years of developing software interfaces for them. And it is much more akin to the HPC model, where high-performance HPC users, or really, the superusers, want to understand everything about the bare metal to optimize their algorithms to it. That's still a model that is the most performant in quantum computing, so things like Braket, things like Azure are a level of abstraction above that, or a few levels, that make it hard to do that right now. But I think their model, their method, if they can work to close that gap and if the hardware providers can work to obscure away the necessity of some of these bare-metal understandings — that whole chain will become more performant and will probably win out over time, in my view.

MN: Got it. Thanks. If I could pivot to IBM specifically, you have some experience working at IBM yourself, and do you think IBM's lead... I believe, in some metrics, they are the leader in terms of quantum computers installed around the world. Could you address whether you think they're in a leadership position? And if so, is that lead durable, or is it primarily a roadmap and marketing lead? I wonder what you think of their positioning right now.

Expert: I will say that I think the advantage that they have from having Qiskit and the IBM Cloud being the first available software SDK for Quantum, and the first available quantum computer online on the cloud... I think that lead is extremely hard for anyone else to overcome. In many ways, Qiskit is the default programming language for quantum computers, and certainly the

interfaces that underlie it are.

Expert: it was a lot of effort from IBM to cement this, and to develop all this software that's usable by the community, and I think that's going to be very hard for others to overcome, because everyone is playing catch-up, basically. People at Google have told me, well, you know, once corner computers are ready, we're just going to put them on the cloud, that's easy, that's trivial for us, right? And I think that what people internal ie. us inside of IBM have learned is, no, that's not trivial. It's very hard to make these things useful and easy for non experts inside to use them. So just from that perspective, I think IBM has a tremendous advantage. It's going to be difficult for other companies to overcome. I think that leadership has also allowed them, basically, the best pick of hiring for a very, very long time. So that, you know, the power, the real technical advantage IBM has is not only the leadership, but the quality from top to bottom. By being able to hire the smartest scientists, or the best scientists. And so that leads, a lot of the innovations and new deliverables that come out are from that, right? Now, to be fair, you know, many other companies also have lots of good talent. But it's the conglomeration that has happened at IBM is to some extent, unique.

But I think overall what we're seeing now is that their hardware is being outpaced by the trapped ions, by the neutral atoms, by even some superconductors, along certain metrics. So we're certainly seeing that they're no longer the best provider of quantum computers, in terms of the quality of the actual computers themselves. And that's going to put them at a bit of a disadvantage moving forward, especially, I keep mentioning this early fault-tolerant era, but that's where we're really going to see trapped ions and neutral atoms, in my view, start to take the lead in terms of quality of demonstrations and applications, right? And along these signposts. So that starts to get more and more momentum, then it's going to be an interesting question of whether or not the usability and adaptability of IBM's wins out over the quality of the others. And I do think we'll see this bifurcation in a way. The people who care about that quality, right, and need the high-performance solutions are going to be working with the trapped ions and neutral atoms. But they're not going to have the easy accessibility to their hardware, so they're going to be much tighter integration. Or it's going to have to be a smaller subset of the market. And I think we'll still see a lot of market capture by IBM, for sure, but they may not be making the headlines in terms of the best demonstrations in the next little while.

MN: And what do you think about IBM's roadmap credibility, specifically, its track record of hitting milestones, and do you think they're... I mean, they've got this very long-term roadmap they've published. Do you think that's realistic?

Expert: I mean, I think no roadmap is realistic, right? I think, we should keep in mind that the roadmaps are best guesses, obviously, as to where the technology's gonna go, but they're as much targets for the internal development teams as they are public-facing information, and I think in particular, IBM, I mean, I can't say for other companies, because I haven't worked for them, but for IBM, you know, they're very transparent about the internal roadmap and the external roadmap are more or less the same, at least they used to be when I was there. So they are very ambitious. And I think overall, you know, some percent, some part of every deliverable is typically there. And I think what a good thing about IBM is that they're very quick to pivot if needed. So you see roadmaps change very rapidly because it's been realized that approaches didn't work, for instance. And you see that with other companies, too. IonQ is pivoting very hard onto different platforms for their ion traps, because they identify that their initial plan wasn't going to be scalable. So I think that's one thing to keep in mind overall, with the quantum computer companies, it's almost worse if a roadmap doesn't change year to year, because it means that in some sense, nothing new was learned, and that's the worst possible situation for a quantum computing company. Because if everything is working, either your approach is not going to work long-term, or something is wrong at a different kind of question.

MN: I'm going to pivot to some questions from the audience. I've got a ton of questions, so I'll try to get through as many as I can here. So, firstly, can multiple workloads be run at the same time on the same system, or does everything have to be run sequentially?

Expert: Absolutely, that is possible. Right now, as far as I know, no cloud service enables it, mostly for a data security question. So this is actually a whole big field in itself, is if you had a big quantum computer and you're running multiple workloads on it, how do you keep someone from being, like, a nefarious actor from using their workload to spy on the other person's workload. And ways to secure that is a very, very difficult problem cryptographically, because quantum is involved. So at the moment, no provider enables that, because it is dangerous in principle, even though, to be fair, no one's doing sensitive stuff right now, on Cloud Quantum.

MN: And once quantum supremacy is achieved, what are the implications for the ongoing rates of change of quantum capabilities?

Expert: I think we will see a very rapid adoption of algorithm research, where people are going to look for ways to transplant whatever algorithm was able to demonstrate quantum advantage to other application domains. Moving, say, a time series

prediction algorithm from market forecasting to Navier-Stokes solutions for nonlinear dynamics of fluids. So, I think we'll see a lot of that research become extremely popular, both industrially and academically. I don't know, I don't think it will necessarily change the path, the speed of hardware development.

MN: Because.

Expert: That really doesn't depend on the demonstration so much as it depends on how the individual components are being put together, and how the system is being architected.

MN: Great question here. In classical computing, there are many common components that allow the industry to scale growth and margins rapidly. Are there any shared components between the different quantum architectures?

Expert: Yes, and that's a great question, because it brings up the importance of all the development of the supporting technologies. So for things like ions and atoms, lasers are a huge important component, because there's lots of laser control. Modulators for optics. All of these optical components are very important and shared as well. Photon detectors, those are hugely, hugely important. Across photonics and atoms and ions. For the manufactured qubits, which normally operate at RF and microwave, or microwave frequencies, you're looking at microwave signal generators, which are these big classical control boxes I was talking about, the quality of control lines of cabling, of isolators and circulators, and then the long-term stretch goals of these cryo-CMOS that you can operate at 4 Kelvin or below, or this single flux quantum control. So there's a tremendous amount of shared technology in the control stack. And that is actually where the bulk of the cost is, and the bulk of the improvement needs to be made to reduce cost and improve deployability.

MN: There's a question here, I'm not sure I fully understand it, but, a follow-up: did you say it would take \$100,000 of electricity to break one cryptographic code, or perhaps they heard this from somewhere else? Is quantum computing energy efficient?

Expert: I did say that, and the answer is, well, I mean, if it's energy efficient, it kind of depends on your perspective, right? I think, absolutely, we should say it's not, in the sense of that it is comparable to these HPC data center scale operations. So it's a huge amount of climate impact, it's a huge amount of energy impact, right? So it's important to conceptualize that the problems you want to solve should have an equivalent benefit in some way. Because otherwise, you're burning money, and you're burning power, and you're burning carbon. We're hoping to improve all of that, and a lot of those technologies I just mentioned are ways to minimize that cost budget. But right now, those large-scale, fault-tolerant operations for code-breaking, for high-performance chemistry and biology, those are not cheap in any sense to operate. So their outcomes better be truly transformative.

MN: A slightly different topic: quantum sensing. We didn't really talk about quantum sensing specifically, so can we talk about that? The question was, how might quantum sensing advances affect sampling rates for machines using manufactured qubits, or any other comments you want to make about quantum sensing?

Expert: That's an interesting question. So what we do know is that, quantum sensing-like approaches, their impact in quantum computing, isn't going to be, what we could say, exponential. It's not going to be able to orders and orders of magnitude improve performance of sampling rates. It can only do it polynomially. This is a mathematical fact. It's very hard, it's unclear that the technical challenges that need to be overcome are worth it in the long run. Sensing itself, sure, tremendous amount of applications, potential for gravitational wave astronomy, for mining, detecting ground deposits, for mapping the Earth's magnetic field, and we're seeing companies that are doing this right now. They're not doing true quantum sensing, in the sense they're not using the quantum features of their sensors, but they're using sensors that are as good as they are because we were building qubits out of them before. And they've transplanted all that knowledge and expertise of controlling the qubits into a classical sensor built from that qubit. There's quite a few examples, especially in magnetic field sensing, where people are starting to deploy these in the real world.

MN: Great, and just one last question, because we're almost top of the hour. How well poised is software and cloud services to fulfill quantum computing demand across applications once the hardware is ready, or fault-tolerant, or coherent enough for commercially useful quantum advantage?

Expert: Far from being where it needs to be. In closing that gap of allowing someone who's a software developer or an application-focused person, you know, even a PhD expert in chemistry or material science to be able to program and use a quantum computer in a highly performant way, the gap is still tremendous. You basically need a quantum computing expert like myself or an applications-focused person on the call with them to enable that implementation to be truly performant and well done. So I think we're very far away from the plug-and-play model, and there's a lot of discussions that have to happen

on figuring out what is the best way to do this. How do you enable somebody who doesn't understand anything about this computational model to have the tools they need to be able to use it efficiently? That's still in the planning stages, really. Most of the people are working on this, people have their ideas, but it's yet to be seen what ends up being the best way to go about this.

MN: Okay, well, let's... I think it's time to wrap up. Thanks very much, Expert, for joining us today. Really greatly appreciate it.

Expert: It was great chatting with you, and I enjoyed all the questions. Thanks again.

MN: If you have any more questions, unfortunately there's not going to be a replay of this, but if you have any questions, feel free to reach out to me at mark.newman@bernsteinsg.com. Thanks very much, have a great day.

Expert: Take care, everyone.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein's "affiliates" relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY

This research publication covers six or more companies. For valuation methodology and other company disclosures:

Please visit: <https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action>.

Or, you can also write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

RISKS

This research publication covers six or more companies. For risks and other company disclosures:

Please visit: <https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action>.

Or, you can also write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 500 Banks And Financial Services Index (BEBANKS) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 500 Insurance Index (BEINSUR) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, and the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- Credit Outperform (C-OP): The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

- **Credit Neutral (C-N):** The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Underperform (C-UP):** The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.1%	16.5%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	36.3%	17.8%
Underperform	SELL	12.6%	14.9%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.

As of March 31, 2026. All figures are updated quarterly.

Prior to April 1, 2024, Sanford C. Bernstein & Co., LLC. issued the ratings and price target information in the graph(s) below for the following companies: Apple Inc, Dell Technologies Inc, Hewlett Packard Enterprise Co, HP Inc and International Business Machines Corp.

PRICE CHARTS/ RATINGS AND PRICE TARGET HISTORY

This research publication covers six or more companies. For price chart and other company disclosures, please visit <https://bernstein-autonomous.bluematrix.com/sellside/Disclosures.action> or you can write to the Director of Compliance, Bernstein Institutional Services LLC, 245 Park Avenue, New York, NY 10167.

CONFLICTS OF INTEREST

Bernstein and/or affiliates have received compensation for investment banking services in the past twelve months from Hewlett Packard Enterprise Co, HP Inc and International Business Machines Corp.

Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Apple Inc.

Bernstein and/or affiliates have received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Apple Inc, Dell Technologies Inc, Hewlett Packard Enterprise Co, HP Inc and International Business Machines Corp.

An affiliate of Bernstein has received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: HP Inc and Western Digital Corp.

Bernstein and/or affiliates expect to receive or intend to seek compensation for investment banking services in the next three months from Hewlett Packard Enterprise Co.

Certain affiliates of Bernstein act as market maker or liquidity provider in the equities securities of: Apple Inc, Dell Technologies Inc, Hewlett Packard Enterprise Co, HP Inc and International Business Machines Corp.

Certain affiliates of Bernstein act as market maker or liquidity provider in the debt securities of: Dell Technologies Inc, Hewlett Packard Enterprise Co and International Business Machines Corp.

Affiliates of Bernstein managed or co-managed in the past twelve months a public offering of securities of Hewlett Packard Enterprise Co, HP Inc and International Business Machines Corp.

Bernstein and/or affiliates had an investment banking client relationship during the past twelve months with Hewlett Packard Enterprise Co, HP Inc and International Business Machines Corp.

OTHER MATTERS

The legal entity(ies) employing the analyst(s) listed in this report, and their location, can be determined by the country code of their phone number, as follows:

- +1 Bernstein Institutional Services LLC; New York, New York, USA
- +44 Bernstein Autonomous LLP; London UK
- +212 Société Générale Africa Technologies & Services; Casablanca, Morocco
- +33 BSG France S.A.; Paris, France
- +34 BSG France S.A.; Madrid, Spain
- +41 Bernstein Autonomous LLP; Geneva, Switzerland
- +49 BSG France S.A.; Frankfurt, Germany
- +91 Sanford C. Bernstein (India) Private Limited; Mumbai, India
- +852 Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司; Hong Kong, China
- +65 Sanford C. Bernstein (Singapore) Private Limited; Singapore
- +81 Sanford C. Bernstein Japan KK; Tokyo, Japan

Where this report has been prepared by research analyst(s) employed by a non-US affiliate, such analyst(s), is/are (unless otherwise expressly noted below) not registered as associated persons of Bernstein Institutional Services LLC or any other SEC-registered broker-dealer and are not licensed or qualified as research analysts with FINRA. Accordingly, such analyst(s) may not be subject to FINRA's restrictions regarding (among other things) communications by research analysts with a subject company, interactions between research analysts and investment banking personnel, participation by research analysts in solicitation and marketing activities relating to investment banking transactions, public appearances by research analysts, and trading securities held by a research analyst account.

Where this report has been prepared by research analyst(s) employed by Société Générale Africa Technologies & Services (part of the Société Générale group of companies), it has been prepared on behalf of a Bernstein company under a Global Services Agreement in place between Bernstein and Société Générale.

CERTIFICATION

Each research analyst listed in this report, who is primarily responsible for the preparation of the content of this report, certifies that all of the views expressed in this publication accurately reflect that analyst's personal views about any and all of the subject securities or issuers and that no part of that analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views in this publication.

II. ADDITIONAL GLOBAL CONFLICT DISCLOSURES

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e., the private side)

within the Firm, and into other areas, units, groups or affiliates (i.e., public side) of the Firm.

III. OTHER IMPORTANT INFORMATION AND DISCLOSURES

Separate branding is maintained for “Bernstein” and “Autonomous” research products.

- Bernstein produces a number of different types of research products including, among others, fundamental analysis and quantitative analysis under both the “Autonomous” and “Bernstein” brands. Recommendations contained within one type of research product may differ from recommendations contained within other types of research products, whether as a result of differing time horizons, methodologies or otherwise. Furthermore, views or recommendations within a research product issued under one brand may differ from views or recommendations under the same type of research product issued under the other brand. The Research Ratings System for the two brands and other information related to those Rating Systems are included in the previous section.
- Autonomous operates as a separate business unit within the following entities: Bernstein Institutional Services LLC, Bernstein Autonomous LLP, Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 and Sanford C. Bernstein (India) Private Limited. For information relating to “Autonomous” branded products (including certain Sales materials) please visit: www.autonomous.com. For information relating to Bernstein branded products please visit: www.bernsteinresearch.com.

Analysts are compensated based on aggregate contributions to the research franchise as measured by account penetration, productivity and proactivity of investment ideas. No analysts are compensated based on performance in, or contributions to, generating investment banking revenues.

This report has been produced by an independent analyst as defined in Article 3 (1)(34)(i) of EU 596/2014 Market Abuse Regulation (“MAR”) and the same article of MAR as it forms part of United Kingdom domestic law by virtue of the European Union (Withdrawal) Act 2018.

To our readers in the United States: Bernstein Institutional Services LLC, a broker-dealer registered with the U.S. Securities and Exchange Commission (“SEC”) and a member of the U.S. Financial Industry Regulatory Authority, Inc. (“FINRA”) is distributing this publication in the United States and accepts responsibility for its contents. Where this material contains an analysis of debt product(s), such material is intended only for institutional investors and is not subject to the US independence and disclosure standards applicable to debt research prepared for retail investors.

Bernstein Institutional Services LLC may act as principal for its own account or as agent for another person (including an affiliate) in sales or purchases of any security which is a subject of this report. This report does not purport to meet the objectives or needs of any specific individuals, entities or accounts.

To our readers in Canada: If this publication pertains to a Canadian domiciled company, it is being distributed in Canada by Sanford C. Bernstein (Canada) Limited, which is licensed and regulated by the Canadian Investment Regulatory Organization. If the publication pertains to a non-Canadian domiciled company, it is being distributed by Bernstein Institutional Services LLC, which is licensed and regulated by both the SEC and FINRA, into Canada under the International Dealers Exemption.

This document may not be passed onto any person in Canada unless that person qualifies as “permitted client” as defined in Section 1.1 of NI 31-103.

To our readers in Brazil: This report has been prepared by Bernstein Institutional Services LLC, and Banco BTG Pactual S.A. (“BTG”) is responsible for the distribution of this report in Brazil.

To readers in the United Kingdom: This publication has been issued or approved for issue in the United Kingdom by Bernstein Autonomous LLP, authorised and regulated by the Financial Conduct Authority and located at 60 London Wall, London EC2M 5SH, +44 (0)20-7170-5000. Registered in England & Wales No OC343985.

This document is for distribution only to persons who (i) have professional experience in matters relating to investments falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the “Financial Promotion Order”), (ii) are persons falling within Article 49(2)(a) to (d) (“high net worth companies, unincorporated associations, etc.”) of the Financial Promotion Order, (iii) are outside the United Kingdom, or (iv) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the FSMA) in connection with the issue or sale of any securities may otherwise lawfully be communicated or caused to be communicated (all such persons together being referred to as “relevant persons”). This document is directed only at relevant persons and must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is available only to relevant persons and will be engaged in only with relevant persons.

To our readers in the member states of the EEA: This publication is being distributed by BSG France SA, which is authorised and regulated by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and Autorité des Marchés Financiers (AMF).

To our readers in Hong Kong: This publication is being distributed in Hong Kong by Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, which is licensed and regulated by the Hong Kong Securities and Futures Commission (Central Entity No. AXC846) to carry out Type 4 (Advising on Securities) regulated activities and subject to the licensing conditions mentioned in the SFC Public Register (<https://www.sfc.hk/publicregWeb/corp/AXC846/details>). This publication is solely for professional investors, as defined in the Securities and Futures Ordinance (Cap. 571). The purpose of this report is solely to provide an analysis of the issuers referred to in this report and is not intended for any purpose contrary to the laws of Hong Kong.

To our readers in Singapore: This publication is being distributed in Singapore by Sanford C. Bernstein (Singapore) Private Limited, only to accredited investors or institutional investors, as defined in the Securities and Futures Act 2001 of Singapore ("SFA"). Recipients in Singapore should contact Sanford C. Bernstein (Singapore) Private Limited in respect of matters arising from, or in connection with, this publication. Sanford C. Bernstein (Singapore) Private Limited is regulated by the Monetary Authority of Singapore and licensed under the SFA as a capital markets services licence holder for dealing in capital markets products that are securities and collective investment schemes and an exempt financial adviser for advising on, issuing and promulgating analyses and reports on securities. Sanford C. Bernstein (Singapore) Private Limited is registered in Singapore with Company Registration No. 20213710W and located at One Raffles Quay, #27-11 South Tower, Singapore 048583, +65-6230-4612.

To our readers in the People's Republic of China: The securities referred to in this document are not being offered or sold and may not be offered or sold, directly or indirectly, in the People's Republic of China (for such purposes, not including the Hong Kong and Macau Special Administrative Regions or Taiwan, the "PRC") in contravention of any applicable laws of the PRC.

This document does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC to any person to whom it is unlawful to make the offer or solicitation in the PRC.

We do not represent that this document may be lawfully distributed, or that any securities may be lawfully offered, in compliance with any applicable registration or other requirements in the PRC, or pursuant to an exemption available thereunder, or assume any responsibility for facilitating any such distribution or offering. In particular, no action has been taken by us which would permit a public offering of any securities or distribution of this document in the PRC. Accordingly, the securities are not being offered or sold within the PRC by means of this document or any other document. Neither this document nor any advertisement or other offering material may be distributed or published in the PRC, except under circumstances that will result in compliance with any applicable laws and regulations.

To our readers in Japan: This publication is being distributed in Japan by Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社), which is registered in Japan as a Financial Instruments Business Operator with the Kanto Local Finance Bureau (registration number: The Director-General of Kanto Local Finance Bureau (FIBO) No.3387) and regulated by the Financial Services Agency. It is also a member of Japan Investment Advisers Association. This publication is solely for qualified institutional investors in Japan only, as defined in Article 2, paragraph (3), items (i) of the Financial Instruments and Exchange Act.

For the institutional client readers in Japan who have been granted access to the Bernstein website by Daiwa Securities Group Inc. ("Daiwa"), your access to this document should not be construed as meaning that Bernstein is providing you with investment advice for any purposes. Whilst Bernstein has prepared this document, your relationship is, and will remain with, Daiwa, and Bernstein has neither any contractual relationship with you nor any obligations towards you.

To our readers in Australia: Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is responsible for distributing research in Australia. It is regulated by the Securities and Exchange Commission under U.S. laws, by the Financial Conduct Authority under U.K. laws, which differs from Australian laws. Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is exempt from the requirement to hold an Australian financial services license under the Corporations Act 2001 in respect of the provision of the following financial services to wholesale clients:

- providing financial product advice;
- dealing in a financial product;
- making a market for a financial product; and
- providing a custodial or depository service.

To our readers in India: This publication is being distributed in India by Sanford C. Bernstein (India) Private Limited (SCB India) which is licensed and regulated by Securities and Exchange Board of India ("SEBI") as a research analyst entity under the SEBI (Research Analyst) Regulations, 2014, having registration no. INH000006378 and as a stock broker having registration no. INZ000213537. SCB India is currently engaged in the business of providing research and stock broking services. Please refer to

www.bernsteinresearch.in for more information.

- SCB India is a Private limited company incorporated under the Companies Act, 2013, on April 12, 2017 bearing corporate identification number U65999MH2017FTC293762, and registered office at Level 3A, 4th Floor, First International Financial Centre, Plot Nos C-54 and C-55, G Block, Near CBI Office, Bandra Kurla Complex, Bandra (East), Mumbai 400098, Maharashtra, India (Phone No: +91-22-68421401).
- For details of Associates (i.e., affiliates/group companies) of SCB India, kindly email MUM-BERNSTEIN-InCompliance@bernsteinsg.com.
- SCB India does not have any disciplinary history as on the date of this report.
- Except as noted above, SCB India and/or its Associates (i.e., affiliates/group companies), the Research Analysts authoring this report, and their relatives
 - do not have any financial interest in the subject company
 - do not have actual/beneficial ownership of one percent or more in securities of the subject company;
 - is not engaged in any investment banking activities for Indian companies, as such;
 - have not managed or co-managed a public offering in the past twelve months for any Indian companies;
 - have not received any compensation for investment banking services or merchant banking services from the subject company in the past 12 months;
 - have not received compensation for brokerage services from the subject company in the past twelve months;
 - have not received any compensation or other benefits from the subject company or third party related to the specific recommendations or views in this report; and
 - do not currently, but may in the future, act as a market maker in the financial instruments of the companies covered in the report.
 - do not have any conflict of interest in the subject company as of the date of this report.
- Except as noted above, the subject company has not been a client of SCB India during twelve months preceding the date of distribution of this research report. Neither SCB India nor its Associates (i.e., affiliates/group companies) have received compensation for products or services other than investment banking, merchant banking or brokerage services from the subject company in the past twelve months.
- The principal research analyst(s) who prepared this report, members of the analysts' team, and members of their households are not an officer, director, employee or advisory board member of the companies covered in the report.
- Our Compliance officer / Grievance officer is Ms. Rupal Talati, who can be reached at +91-22-68421451, or MUM-BERNSTEIN-InCompliance@bernsteinsg.com / Scbin-investorgrievance@bernsteinsg.com
- The Research investor charter and Terms & Conditions of SCB India are available on its website and may be accessed at [Sanford C. Bernstein \(India\) Private Limited](https://bernsteinresearch.in/) (<https://bernsteinresearch.in/>) for your reference.
- Disclaimer: Registration granted by SEBI, and certification from NISM, is in no way a guarantee of performance of the intermediary or provide any assurance of returns to investors. Investments in securities market are subject to market risks. Read all the related documents carefully before investing.

To our readers in Switzerland: This document is provided in Switzerland by or through Bernstein Autonomous LLP, and is provided only to qualified investors as defined in article 10 of the Swiss Collective Investment Scheme Act ("CISA") and related provisions of the Collective Investment Scheme Ordinance and in strict compliance with applicable Swiss law and regulations. The products mentioned in this document may not be suitable for all types of investors. This document is based on the Directives on the Independence of Financial Research issued by the Swiss Bankers Association (SBA) in January 2008.

To our readers in the Middle East: Bernstein Autonomous LLP, DIFC branch has its principal office at Gate Village 06, DIFC, Dubai, UAE. Bernstein Autonomous LLP, DIFC branch is regulated by the Dubai Financial Services Authority (DFSA) with the registration number CL10040 and is provisioned for Arranging Deals in Investments and Advising on Financial Products. All communications and services are directed at Professional Clients and Market Counterparties only (as defined in the DFSA rulebook). Persons other than Professional Clients and Market Counterparties, such as Retail Clients, are not the intended recipients of our communications or services.

LEGAL

All research publications are disseminated to our clients through posting on the firm's password protected websites, bernsteinresearch.com and autonomous.com. Certain, but not all, research publications are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience.

This publication has been published and distributed in accordance with the Firm's policy for management of conflicts of interest in investment research, a copy of which is available from Bernstein Institutional Services LLC, Director of Compliance, 245 Park Avenue, New York, NY 10167. Additional disclosures and information regarding Bernstein's business are available on our website www.bernsteinresearch.com.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. This publication is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of, or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or which would subject any of the entities referenced herein or any of their subsidiaries or affiliates to any registration or licensing requirement within such jurisdiction. This publication is based upon public sources we believe to be reliable, but no representation is made by us that the publication is accurate or complete. We do not undertake to advise you of any change in the reported information or in the opinions herein. This publication was prepared and issued by entity referred to herein for distribution to eligible counterparties or professional clients. This publication is not an offer to buy or sell any security, and it does not constitute investment, legal or tax advice. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with their professional advisors in light of their specific circumstances. The value of investments may fluctuate, and investments that are denominated in foreign currencies may fluctuate in value as a result of exposure to exchange rate movements. Information about past performance of an investment is not necessarily a guide to, indicator of, or assurance of, future performance.

This report is directed to and intended only for our clients who are "eligible counterparties", "professional clients", "institutional investors" and/or "professional investors" as defined by the aforementioned regulators, and must not be redistributed to retail clients as defined by the aforementioned regulators. Retail clients who receive this report should note that the services of the entities noted herein are not available to them and should not rely on the material herein to make an investment decision. The result of such act will not hold the entities noted herein liable for any loss thus incurred as the entities noted herein are not registered/ authorised/ licensed to deal with retail clients and will not enter into any contractual agreement/arrangement with retail clients. This report is provided subject to the terms and conditions of any agreement that the clients may have entered into with the entities noted herein. All research reports are disseminated on a simultaneous basis to eligible clients through electronic publication to our client portal.

The information in this report was prepared by Bernstein solely for the internal business use of our clients. Clients may store, display, analyze, reformat and print the information in this report for this limited use only. Clients may not copy, alter, create derivative works, resell, reverse engineer, commercially exploit, share or distribute any part of the information contained herein for any purpose without Bernstein's express written consent. These restrictions include extracting data or using the content to develop indices or other products. Further, you may not use this report, or any portion of this report, to train or finetune any third-party machine learning or artificial intelligence system, or as a prompt or input into any such system. You also may not, without Bernstein's express written consent, do any of the foregoing in connection with your own internal machine learning or artificial intelligence system.

Bernstein may use artificial intelligence tools in the preparation of its materials. Any such materials are reviewed by Bernstein's research analysts prior to publication.

This report has been prepared for information purposes only and is based on current public information that we consider reliable, but the entities noted herein do not warrant or represent (express or implied) as to the sources of information or data contained herein are accurate, complete, not misleading or as to its fitness for the purpose intended even though the entities noted herein rely on reputable or trustworthy data providers, it should not be relied upon as such. Opinions expressed are the author(s)' current opinions as of the date appearing on the material only and we do not undertake to advise you of any change in the reported information or in the opinions herein.

Any references to Societe Generale herein are purely factual, based upon publicly available information, and included for comparative purposes only. They do not constitute an opinion or recommendation with respect to the securities of Societe Generale.

This publication was prepared and issued by the entity referred to herein for distribution to eligible counterparties or professional clients. The information in this report is intended for general circulation and does not constitute an offer to buy or sell any security, investment, legal or tax advice nor a personal recommendation, as defined by any of the aforementioned regulators. It does not take into account the particular investment objectives, financial situations, or needs of individual investors. The report has not been reviewed by any of the aforementioned regulators and does not represent any official recommendation from the aforementioned regulators. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with advice sought from a financial adviser regarding the suitability of the investment product, taking into account the specific investment objectives, financial situation or particular needs of any recipient of the recommendation, before the recipient makes a commitment to purchase the investment product.

The analysis contained herein is based on numerous assumptions. Different assumptions could result in materially different results. The information in this report does not constitute, or form part of, any offer to sell or issue, or any offer to purchase or subscribe for shares, or to induce engagement in any other investment activity. The value of any securities or financial instruments mentioned in this report may fluctuate subject to market conditions. Information about past performance of an investment is not necessarily a guide to, indicative of, or assurance of future performance. Estimates of future performance mentioned by the research analyst in this report are based on assumptions that may not be realized due to unforeseen factors like market volatility/fluctuation. In relation to securities or financial instruments denominated in a foreign currency other than the clients' home currency, movements in exchange rates will have an effect on the value, either favorable or unfavorable. Before acting on any recommendations in this report, recipients should consider the appropriateness of investing in the subject securities or financial instruments mentioned in this report and, if necessary, seek independent professional advice.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject the entities noted herein to any regulation or licensing requirement within such jurisdiction.

Source: Bloomberg Index Services Limited. BLOOMBERG® is a trademark and service mark of Bloomberg Finance L.P. and its affiliates (collectively "Bloomberg"). Bloomberg or Bloomberg's licensors own all proprietary rights in the Bloomberg Indices. Neither Bloomberg nor Bloomberg's licensors approves or endorses this material, or guarantees the accuracy or completeness of any information herein, or makes any warranty, express or implied, as to the results to be obtained therefrom and, to the maximum extent allowed by law, neither shall have any liability or responsibility for injury or damages arising in connection therewith.

No part of this material may be reproduced, distributed or transmitted or otherwise made available without prior consent of the entities noted herein. Copyright Bernstein Institutional Services LLC Bernstein Autonomous LLP, BSG France S.A., Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited and Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社). All rights reserved. The trademarks and service marks contained herein are the property of their respective owners. Any unauthorized use or disclosure is strictly prohibited. The entities noted herein may pursue legal action if the unauthorized use results in any defamation and/or reputational risk to the entities noted herein and research published under the Bernstein and Autonomous brands.